Douglas R. Bergman

University of Utah
Department of Physics and Astronomy
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bergman@physics.utah.edu (e-mail)

EDUCATION

Yale University, New Haven, CT, Ph.D. in physics, July 1997.

Yale University, New Haven, CT, M.Phil. in physics, May 1991.

Yale University, New Haven, CT, M.S. in physics, Dec. 1990.

University of Chicago, Chicago, IL, A.B. in physics, June 1989.

EMPLOYMENT

Professor, Univ. of Utah	July 2022 – pres.
Associate Professor, Univ. of Utah	Aug 2009 – June 2022
Assistant Professor, Rutgers Univ.	July 2005 – June 2009
Research Assistant Professor, Rutgers Univ.	July 2004 – June 2005
Research Associate, Rutgers University	July 2000 – June 2004
Postdoctoral Fellow, Rutgers University	July 1997 – July 2000
Graduate Research Assistant, Yale University	Feb. 1991 – July 1997
Math and Science Tutor, Pierson Coll., Yale	Sept. 1991 – May 1995
Graduate Teaching Fellow, Yale University	Sept. 1989 – May 1997
Undergraduate Research Assistant, Chicago	Jan. 1988 – Aug. 1989
Teaching Assistant, University of Chicago	Oct. 1986 – Mar. 1987
Summer Intern, General Electric, Cleveland	June – Sept. 1986

RESEARCH

University of Utah, Salt Lake City, UT — Jul. 2025 – pres.
Member of the Trinity Collaboration.

University of Utah, Salt Lake City, UT — Mar. 2018 – pres.
Member of the nuSpaceSim collaboration, "Modeling of the Air Shower Signals from Cosmic Neutrinos for Space-based Experiments".

- Contributed to the Cherenkov radiation from air showers simulation.

University of Utah, Salt Lake City, UT — Feb. 2017 – pres.
Member of the Probe Of Extreme Multi-messenger Astrophysics (POEMMA) Collaboration.

- Convener of the Focal Surface Group.

University of Utah, Salt Lake City, UT Aug. 2011 – pres.

Development of the Non-Imaging Cherenkov (NICHE) array as a low-energy extension of TA & TALE.

University of Utah, Salt Lake City, UT Aug. 2009 – pres.

Member of the Telescope Array (TA) and TA Low Energy Extension (TALE) Collaborations.

- Helped build and deploy the TALE Fluorescence Detector.
- Performed a stereo measurement of the UHECR nuclear composition.
- Performed the first analysis of the UHECR spectrum using monocular fluorescence data in TA.

Rutgers University, Piscataway, NJ Sep. 2004 – Jun. 2009

Member of the Telescope Array (TA) and TA Low Energy Extension (TALE) Collaborations.

- TA Fluorescence Detector Analysis Coordinator.
- Built, deployed and tested a prototype TALE Tower Detector mirror.

Rutgers University, Piscataway, NJ Jun. 2002 – 2008

Member of the FLASH Collaboration (E-165 at SLAC), an experiment to measure to fluorescence yield of electrons in air.

Rutgers University, Piscataway, NJ Jan. 1999 – 2009

Member of the High Resolution Fly's Eye Experiment.

- Observed the GZK Cutoff in the UHECR flux spectrum.
- Measured the UHECR composition using the distribution of shower maxima in the atmosphere.
- Fit the measured HiRes UHECR spectra to a number of different source models.
- Developed phenomenological model of UHECR propagation to determine the expected shape of the UHECR spectrum from extragalactic sources.
- Measured the spectrum of UHECR's in a monocular analysis using time-binned FADC data.
- Developed analysis and visualization programs to work with FADC data.
- Surveyed PMT pointing directions using CCD camera to view star images on a screen at the PMT plane.
- Surveyed mirror positions using differential GPS unit.

Rutgers University, Piscataway, NJ July 1997 – Dec. 2001

Member of the KTeV Experiment: Analyzed data from the 1997 run of KTeV (E832/E799) at the Fermilab Tevatron, measuring the branching ratio and photon spectrum for radiative K_{e3} decays.

Rutgers University, Piscataway, NJ July 1997 – June 1999

Worked on the proposal for the CP/T experiment, an experiment to study K_L-K_S interference immediately downstream of a K^0 production target. Specifically worked on determining the systematic errors involved in measuring x, the $\Delta S = -\Delta Q$ amplitude in K_{e3} decays.

Yale University, New Haven, CT Aug. 1995 – July 1997

Analyzed data from the 1995 run of E865 at the AGS, looking for the decay, $K^+ \to \pi^+\mu^+e^-$. Wrote and defended Ph.D thesis, *A Search for the Decay* $K^+ \to \pi^+\mu^+e^-$, under the supervision of Professor Michael E. Zeller.

Yale University, New Haven, CT July 1990 – June 1995

Participated in all aspects of the design, construction and installation of the E865 detector apparatus at the AGS.

- Wrote a Monte Carlo simulation of the detector using the GEANT package from CERN.
- Over-saw the mapping of the two large spectrometer magnets used in the experiment. Checked this map for consistency against Maxwell's equations and corrected discrepancies therefrom.
- Installed and helped to design the fast trigger system.
- Participated in the design and prototyping of novel proportional wire chambers using graphite on mylar anode planes.
- Studied the feasibility of looking for CP violation in the decay $K \to \pi\pi\pi$.
- Ran simulations of the low halo beam-line design to determine background rates.

University of Chicago, Chicago, IL June 1988 – Sept. 1989

Assisted in the design of THISTLE, a balloon-borne experiment to measure the relative abundances of nuclear isotopes in cosmic rays. Worked to port Monte Carlo simulation code for these abundances to an IBM PC platform.

University of Chicago, Chicago, IL Jan. – June 1988

Assisted in the construction of modules for the Chicago Air Shower Array, a high-energy cosmic ray detector that was deployed at Dugway Proving Grounds, UT.

General Electric, Nela Park, Cleveland, OH June – Sept. 1986

Summer Intern: Studied the intensity of light from the UV lines of mercury vapor lamps under various conditions, including temperature and vapor pressure. Wrote this work up in an internal company memo.

PRESENTATIONS

†*invited talks*

The Cosmic Ray Event Spectrum in the Knee Region measured by the NICHE Array at Telescope Array, 39th International Cosmic Ray Conference, Geneva, Jul. 2025.

†*Kita no Ameterasu, Telescope Array and the Dawn Goddess of the North*, Colloquium, University of Wisconsin, Madison, March 2025.

†*Testing the Compatibility of the Depth of the Shower Maximum Measurements performed at Telescope Array and the Pierre Auger Observatory*, UHECR 2022, L'Aquilla, Oct. 2022.

†*Highlights of Telescope Array*, TeV Particle Astrophysics 2021 (TeVPA 2021), Chengdu (virtual), Oct. 2021.

†*TA, TALE and TAx4, Latest Results*, 17th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2021), Valencia (virtual), Aug. 2021.

†*TA, TALE and TAx4, Latest Results*, 17th International Conference on Topics in Astroparticle and Underground Physics (TAUP 2021), Valencia (virtual), Aug. 2021.

Telescope Array 10-Year Monocular Spectrum Measurement, 37th International Cosmic Ray Conference, Berlin (virtual), July 2021.

Telescope Array Combined Fit to Cosmic Ray Spectrum and Composition, 37th International Cosmic Ray Conference, Berlin (virtual), July 2021.

Searching for the Sources of the Most Energetic Particles in the Universe, Colloquium, University of Utah, Sept. 2020.

†*Telescope Array: Latest Results and Plans*, TAUP 2019, Toyama, Sept. 2019.

First Results from NICHE and the NICHE-TALE Hybrid Detector, 36th International Cosmic Ray Conference, Madison, July 2019.

Combined Fit of the Spectrum and Composition from Telescope Array, 36th International Cosmic Ray Conference, Madison, July 2019.

TA 10-Year Stereo Composition Measurement, 36th International Cosmic Ray Conference, Madison, July 2019.

The Non-Imaging CHErenkov Array(NICHE): First Analyses, APS April Meeting, Denver, CO, April 2019.

†*NICHE: Air-Cherenkov light observation at the TA site*, UHECR 2018, Paris, France, Oct. 2018.

The Non-Imaging CHErenkov (NICHE) Array: A TA/TALE extension using Cherenkov radiation to measure Cosmic Ray Composition to sub-PeV energies, APS April Meeting, Columbus, OH, April 2018.

j-NICHE: Prototype detectors of a non-imaging Cherenkov array at the Telescope Array site, 35th International Cosmic Ray Conference, Busan, S. Korea, July 2017.

Telescope Array measurement of UHECR composition from stereoscopic fluorescence detection, 35th International Cosmic Ray Conference, Busan, S. Korea, July 2017.

†*Results from Telescope Array and TALE*, STARS2017, Havana, Cuba, May 2017.

NICHE: Using Cherenkov Radiation to Extend Telescope Array to Sub-PeV Energies, APS April Meeting, Salt Lake City, UT, April 2016.

†*Results from Telescope Array and TALE*, HAP Workshop Composition 2015, Karslruhe, Sept. 2015.

Imaging and Non-Imaging Cherenkov Hybrid Reconstruction, 34th International Cosmic Ray Conference, The Hague, July 2015.

The NICHE Array: Status and Plans, 34th International Cosmic Ray Conference, The Hague, July 2015.

The distribution of shower longitudinal profiles widths as measured by Telescope Array in stereo mode, 34th International Cosmic Ray Conference, The Hague, July 2015.

†*Results from Telescope Array*, IceCube Particle Astrophysics Symposium, Madison, May 2015.

†*The Non-Imaging CHErenkov Array (NICHE): A TA/TALE Extension to Measure the Flux and Composition of Very-High Energy Cosmic Rays*, Conference on Ultrahigh Energy Cosmic Rays (UHECR2014), Springdale, UT, Oct. 2014.

†*Cosmic Ray Measurements at the Highest Energies: Results from Telescope Array*, 40th Cospar Scientific Assembly, Moscow, Russia, Aug. 2014.

The Spectrum of UHECRs, Colloquium, University of Utah, Sept. 2013.

†*Telescope Array: Recent Results, Future Plans*, Cosmic Ray Anisotropy Workshop, Madison, WI, Sept. 2013.

TA Spectrum Summary, 33rd International Cosmic Ray Conference, Rio de Janeiro, Brazil, July 2013.

An Efficient Technique for the Reconstruction of Extensive Air Showers using Non-Imaging Cherenkov Measurements (poster), 33rd International Cosmic Ray Conference, Rio de Janeiro, Brazil, July 2013.

The Non-Imaging CHErenkov Array (NICHE): A TA/TALE extension to measure the flux and composition of Very-High Energy Cosmic Rays, APS April Meeting, Denver, CO, April 2013.

† *Telescope Array: Recent Results, Future Plans*, 4th Workshop on Air Shower Detection at High Altitudes, Naples, Italy, Jan. 2013.

† *HiRes and TA Spectrum Measurements*, The International Symposium on Future Directions in UHECR Physics, Geneva, Switzerland, Feb. 2012.

Five Decades of Cosmic Rays, Colloquium, University of Utah, Sept. 2011.

The Energy Spectrum of UHECRs using the Telescope Array Fluorescence Detectors in Monocular Mode, 32nd International Cosmic Ray Conference, Beijing, China, Aug. 2011.

Detecting UHECRs using Cherenkov Light, APS April Meeting, Anaheim, CA, April 2011.

† *HiRes Results, the Final Word (almost)*, Workshop on Hadron-Hadron and Cosmic-Ray Interactions at multi-TeV Energies, ECT* Centre, Trento, Italy, Dec. 2010.

Non-Imaging Cherenkov Detection, Informal Seminar, University of California, Irvine, Irvine CA, Aug. 2010.

† *HiRes UHECR Spectrum Measurements*, Bartol Mini-workshop, University of Delaware, Newark DE, Dec. 2009.

The HiRes Stereo Measurement of the UHECR Spectrum, Fall 2009 Meeting of the Four Corners Section of the APS, Colorado School of Mines, Golden CO, Oct. 2009.

† *UHECR's in the Northern Hemisphere: A Status Report: Recent Results from HiRes*, CCAPP Inaugural Seminar, Ohio State University, Columbus OH, Oct. 2009.

Stereoscopic Measurement of the Flux of Ultra High Energy Cosmic Rays by the High Resolution Fly's Eye, 31st International Cosmic Ray Conference, Łódź, Poland, July 2009.

The Energy Spectrum of UHECR's using the TA Fluorescence Detectors in Monocular Mode, 31st International Cosmic Ray Conference, Łódź, Poland, July 2009.

Why HiRes was Able to Observe the GZK Cutoff, Colloquium, University of Utah, April 2009.

† *Recent Results from HiRes*, 20th Recontres de Blois, Blois, France, May 2008.

† *Observation of the GZK Suppression by HiRes*, International Astrophysics Symposium, Golden, CO, May 2008.

Observation of the GZK Cutoff Using by the HiRes Experiment, 30th International Cosmic Ray Conference, Merida, Mexico July 2007.

The TALE Tower Detector (poster), 30th International Cosmic Ray Conference, Merida, Mexico July 2007.

† *TA/TALE Usage*, Aspen Workshop on Cosmic Ray Physics, Aspen CO, April 2007.

† *Observation of the GZK Cutoff Using the HiRes Detector*, SUSY06, Irvine CA, June 2006.

† *Observation of the GZK Cutoff Using the HiRes Detector*, CRIS 2006, Catania, Italy, May 2006.

Fitting the HiRes Spectra, 29th International Cosmic Ray Conference, Pune, India, Aug. 2005.

UHECR Composition Measurements Using the HiRes-II Detector (poster), 29th International Cosmic Ray Conference, Pune, India, Aug. 2005.

Monocular UHECR Spectrum Measurements from HiRes (poster), 29th International Cosmic Ray Conference, Pune, India, Aug. 2005.

† *Fitting the HiRes Spectra*, Physics at the End of the Galactic Cosmic Ray Spectrum, Aspen, Apr. 2005.

From HiRes to the Telescope Array: A Tale of Hybrid Vigor, Colloquium, University of Delaware, Nov. 2004.

Fitting the HiRes Spectra and Monocular Composition, CRIS 2004, Catania, Italy, May 2004.

The Advent of Cosmic Ray Astronomy, Colloquium, University of Oklahoma, Feb. 2004.

Fits of the HiRes Spectra to Astrophysical Models, 28th International Cosmic Ray Conference, Tsukuba, Japan, Aug. 2003.

Measurement of the Flux of UHE Cosmic Rays by the HiRes Detectors Observing in Monocular Mode, 28th International Cosmic Ray Conference, Tsukuba, Japan, Aug. 2003.

Fits of the HiRes Spectra to Astrophysical Models, 2003 APS April Meeting, Philadelphia, PA, April 2003.

The GZK with HiRes, Invited Seminar, The Pennsylvania State University, Oct. 2002.

The GZK with HiRes, Invited Seminar, Brookhaven National Laboratory, Sep 2002.

The UHECR Spectrum with HiRes, 31st International Conference on High Energy Physics, Amsterdam, July 2002.

The UHECR Spectrum with HiRes, 2002 APS April Meeting, Albuquerque, NM, April 2002.

†*Monocular UHECR Spectra with HiRes*, 2002 Aspen Winter Conference on Ultra High Energy Particles from Space, Jan 2002.

A Monocular Spectrum Analysis Using FADC Timing at HiRes, 27th International Cosmic Ray Conference, Aug 2001.

Determining the Alignment of HiRes Optics Using a CCD Camera, 27th International Cosmic Ray Conference, Aug 2001.

Recent Results from the High Resolution Fly's Eye Experiment, 20th Texas Symposium on Relativistic Astrophysics, Dec. 2000.

†*A New Measurement of the Radiative K^0_{e3} Branching Ratio and Photon Spectrum*, Meson 2000, May 2000.

A Search for the Decay $K^+ \to \pi^+\mu^+e^-$, KTeV Lunch Talk, Aug. 1997

A Search for the Decay $K^+ \to \pi^+\mu^+e^-$, APS Washington Meeting, April 1997

E865: A Status Report, RHIC-AGS Users Meeting, May 1996

FUNDING

NSF PHY-2609404, $2,446,218, Charles Jui, Douglas Bergman, John Matthews, *Base Support of the University of Utah Cosmic Ray Group and Analysis of Telescope Array Data*, Jul. 2026–Jun. 2029. *[pending]*

NSF PHY-2609403, $1,775,596, Charles Jui, Douglas Bergman, John Matthews, *University of Utah Cosmic Ray Operation of Telescope Array TAx4*, Jul. 2026–Jun. 2029. *[pending]*

NSF PHY-2609402, $600,506, Dennis Soldin, Douglas Bergman, RObert Wayne Springer, *Data Analysis and Simulations for the Trinity Demonstrator in Support of Trinity-One Development*, Jul. 2026–Jun. 2029. *[pending]*

NASA 24-APRA24-0042,$184,588, John Krizmanic *et al.*, *Continued Development of nuSpaceSim: Modeling of the Extensive Air Shower Signals from Cosmic Neutrinos for Suborbital and Space-based Experiments*, Oct. 2025–Sep. 2028. *[selectable]*

NASA 21-APRA21-0071, $136,423, John Krizmanic *et al.*, *nuSpaceSim: Modeling of Extensive Air Shower Signals from Cosmic Neutrinos for Space-based Experiments*, Jul. 2022–Jun. 2026.

NSF PHY-2310163, $400,000, Charles Jui, Douglas Bergman, John Matthews, Carsten Rott, *Analysis of Data from the Telescope Array Observatory*, Aug. 2023–Jul. 2026.

NSF PHY-2209584, $310,000, Charles Jui, Douglas Bergman, John Matthews, Carsten Rott, *Analysis of Data from the Telescope Array Observatory*, Oct. 2022–Sep. 2023.

NSF PHY-2209583, $2,300,000, Charles Jui, Douglas Bergman, John Matthews, Carsten Rott, *Operation of the Telescope Array Observatory*, Sept. 2022–Aug. 2025.

Utah Research Incentive Seed Grant, $34,968, John Matthews, Douglas Bergman, *Prototyping Cosmic Ray Detectors to Add to the IceCube Neutrino Observatory*, Apr. 2022–Mar. 2023.

NSF PHY-2112904, $865,000, Charles Jui, Douglas Bergman, Carsten Rott, John Matthews, *Operation of the Telescope Array Observatory*, Sep. 2021–Aug. 2022.

NSF PHY-2012934, $841,026, Charlie Jui, Douglas Bergman, John Matthews, Pierre Sokolsky, University of Utah, *Operation of the Telescope Array, TALE, and the TA×4 Expansion by the University of Utah*, Sep. 2020–Aug. 2021.

NASA 17-APRA17-0066, $107,381, John Krizmanic *et al.*, University of Utah, *Modeling of the Air Shower Signals from Cosmic Neutrinos for Space-based Experiments*, Feb. 2019–Feb. 2023.

NSF PHY-1806797, $2,172,872, Gordon Thomson *et al.*, University of Utah, *Baseline Support of the University of Utah Cosmic Ray Physics Group Including Analysis of Telescope Array, TALE, and TAx4 Data*, July 2018–Jun. 2021.

NSF PHY-1712517, $2,424,080, Gordon Thomson *et al.*, University of Utah, *University of Utah - Operation of the Telescope Array, TALE, and the TAx4 Expansion*, Aug. 2017–Jul. 2020.

NSF PHY-1607727, $1,775,000, Gordon Thomson *et al.*, University of Utah, *Telescope Array TAx4 Upgrade Deployment Proposal*, Aug. 2016–Jul. 2019.

NSF PHY-1404502, $2,311,356, Pierre Sokolsky *et al.*, University of Utah, *Telescope Array Operations and Data Analysis by the University of Utah Cosmic Ray Group*, Sep. 2014–Aug. 2017.

NSF PHY-1404495, $1,846,443, Pierre Sokolsky *et al.*, University of Utah, *Baseline Support of the University of Utah Cosmic Ray Physics Group Including Analysis of the Telescope Array*, Sep. 2014–Aug. 2017.

NSF PHY-1069286, $1,707,240, Pierre Sokolsky *et al.*, University of Utah, *Telescope Array Operations and Data Analysis by the University of Utah Cosmic Ray Group*, Sep. 2011–Aug. 2014.

NSF PHY-1069280, $3,124,772, Pierre Sokolsky *et al.*, University of Utah, *Baseline Support of the University of Utah Cosmic Ray Physics Group Including Analysis of the Telescope Array*, Sep. 2011–Aug. 2014.

NSF PHY-0968374, $400,000, Gordon Thomson & Douglas Bergman, University of Utah, *Research in Ultrahigh Energy Cosmic Ray Physics*, May 2010–Apr. 2011.

NSF PHY-0649681, $1,173,500, Gordon Thomson & Douglas Bergman, Rutgers University, *Experimental Research in Ultrahigh Energy Cosmic Ray Physics*, May 2007–Apr. 2010.

NSF PHY-0305516, $3,474,000, Gordon Thomson, Thomas Devlin, Stephen Schnetzer, Sunil Somalwar, Amitabh Lath, Douglas Bergman and Eva Halkiadakis, Rutgers University, *Experimental Research In Elementary Particle Physics*, Jan 2004–Apr. 2007.

POSTDOCTORAL FELLOWS

Zane Gerber, University of Utah	Aug 2025 – pres.
JiHyun Kim, University of Utah	Oct. 2020 – Aug 2025
Thomas Stroman, University of Utah	Jul. 2010 – Aug. 2015
Lauren Scott, Rutgers University	Jul. 2005 – Jul. 2010

GRADUATE STUDENTS

Sneha Singh, University of Utah	May 2025 – pres.
Zane Gerber, University of Utah	May 2021 – Aug 2025
Isaac Buckland, University of Utah	May 2020 – Dec. 2024
Greg Furlich, University of Utah	Aug. 2015 – May 2020
Jian Lan, University of Utah	Aug. 2012 – Apr. 2016
Saskia Innemee, University of Utah	Aug. 2010 – May 2012
Rhett Zollinger, University of Utah	Aug. 2010 – Apr. 2011
Sean Stratton, Rutgers University	Aug. 2005 – May 2012

UNDERGRAD STUDENTS

Jakub Ziembicki, University of Utah	Apr. 2023 – May 2024.
Faizah Siddique, UMass Amherst (REU)	May 2020 – Aug. 2020
Zane Gerber, University of Utah	Aug. 2018 – Aug. 2019
Connor Houghton, University of Utah	Jan. 2018 – Apr. 2020
Matthew Dutson, University of Utah	Jan. 2016 – Dec. 2017
Jeffrey Rodgers, University of Utah	Jan. 2016 – Apr. 2018
Nathan Waugh, Weber State University (REU)	Jun. 2014 – Jul. 2014
Bingran Wang, University of Utah	Dec. 2012 – Apr. 2015
Saskia Innemee, Rutgers University	Sep. 2008 – May 2010
Oren Levi, Rutgers University	May 2008 – Aug. 2008
Aneta Biesiedecka, Rutgers University	May 2007 – Aug. 2008

TEACHING

University of Utah

- **PHYS 2210: Physics I** S2026
- **PHYS 4010: Physics V, E&M and QM** F2025
- **PHYS 2210: Physics I** S2025

- **PHYS 4010: Physics V, E&M and QM** F2024
- **PHYS 5210: Introduction to Gravity** S2024
- **PHYS 4010: Physics V, E&M and QM** F2023
- **PHYS 7320: Quantum Theory II** S2023
- **PHYS 5110: Nuclear & Particle Physics** F2022
- **PHYS 7320: Quantum Theory II** S2022
- **PHYS 7320: Quantum Theory II** S2021
- **PHYS 5110: Nuclear & Particle Physics** F2020
- **PHYS 7320: Quantum Theory II** S2020
- **PHYS 5210: Introduction to Gravity** F2018
- **PHYS 4420: Classical Physics II** S2018
- **PHYS 4410: Classical Physics I** F2017
- **PHYS 3730: Computing in Physics** (2 sect.) F2016
- **PHYS 3719/3729: Undergrad Lab** S2016
- **PHYS 3610/6610: Electronics I** F2015
- **PHYS 3740: Modern Physics** S2015
- **PHYS 3610/6610: Electronics I** F2014
- **PHYS 3740: Modern Physics** S2014
- **PHYS 3610/6610: Electronics I** F2013
- **PHYS 7910-013: Astroparticle Physics** S2013
- **PHYS 3610/6610: Electronics I** F2012
- **PHYS 3620/6620: Electronics II** S2012
- **PHYS 3610/6610: Electronics I** F2011
- **PHYS 3620/6620: Electronics II** S2011
- **PHYS 3610/6610: Electronics I** F2010
- **PHYS 3620/6620: Electronics II** S2010

Rutgers University

- **PHYS 272: Honors Physics II** S2009
- **PHYS 271: Honors Physics I** F2008
- **PHYS 272: Honors Physics II** S2008
- **PHYS 271: Honors Physics I** F2007
- **PHYS 272: Honors Physics II** S2007
- **PHYS 271: Honors Physics I** F2006
- **PHYS 124: Analytical Physics I-B** S2006
  Recitation Section Leader

SERVICE

Utah, University Committees

- Senate Committee on Academic Freedom and Faculty Rights 2017–2024.

Utah, College Committees (Science)

- College of Science Council 2012–2013

Utah, Departmental Committees (Physics)

- Teaching Assistant Committee, co-chair, 2025–pres.
- Computing Committee, chair, 2025–pres.
- Awards Committee (Student), chair 2026–pres.
- Awards Committee (Student), 2025–pres.
- Awards Committee (Student), chair 2024–2025
- Futures Committee 2024–2025
- Graduate Comprehensive Exam, chair 2023–2025
- Public Presentation Committee 2023–2024
- Graduate Admissions Coordination, chair 2022–2023
- Faculty Search in Multi-messenger Astronomy 2022–2023
- Graduate Program, Curriculum & Recruitment 2022-2024
- Director of Graduate Studies 2021–2022
- Graduate Recruitment & Admissions, chair 2020–2021
- Seminar Series – Virtual 2020–2021
- Graduate Recruitment & Admissions 2018–2022
- Director of Graduate Studies 2018–2019
- Policy Board 2017–2019
- Admissions Committee 2011–2018
- Director of Graduate Studies, elect 2017–2018
- Recruitment Committee 2017–2018
- HEAP Seminar Committee 2014–2018
- Colloquium Committee, chair 2016–2017
- Colloquium Committee 2014–2017
- HEAP Seminar Committee, chair 2015–2016
- Admissions Committee, chair 2012–2015
- Advising Committee (Graduate) 2012–2014
- Public Education & Outreach 2012–2013
- Common Exam Committee 2011–2012
- HEAP Seminar Committee, chair 2011–2012
- Space Committee 2010–2012
- Futures Committee 2010–2011
- HEAP Seminar Committee 2009–2010

Rutgers, Departmental Committees (Physics)

- Society of Physics Students Faculty Liaison 2008–2009
- Undergraduate Studies 2006–2009
- Computer Services 2005–2009
- Graduate Recruitment Committee 2005–2006
- Written Qualifier Grading Committee 2005–2006

Organizing

- International Orgainzing Committee, HAP Workshop Composition 2015, Karslruhe, Germany, Sept. 2015.
- Local Organizing Committee & session chair, UHECR 2014, Springdale UT, Sept. 2014.

Reviewing

- Advances in Space Research (5)
- Astrophysical Journal (1)
- Astroparticle Physics (7)
- JCAP (1)
- NASA APRA-SAT Review Panel
- Nazarbaev University
- Modern Physics Letters A (1)
- National Science Foundation (12)
- Textbook: Bauer & Westfall, *University Physics*.

Coaching

- Science Olympiad, Sounds of Music, Torrey Pines High School 2010–2011

Judging

- North Jersey Regional Science Fair 2005–2006

All Publications of D. R. Bergman

[1] R. U. Abbasi et al. "Searching for EeV photons with Telescope Array Surface Detector and neural networks". In: (Dec. 2025). arXiv: 2512.01638 [astro-ph.GA].

[2] Rasha Abbasi et al. "Study of air shower reconstruction with asymmetric lateral distribution function for TAx4 SD". in: *PoS* ICRC2025 (2025), p. 378. DOI: 10.22323/1.501.0378.

[3] Zane Gerber and Douglas Bergman. "Hybrid Measurement of UHECR Mass Composition with TAx4". In: *PoS* ICRC2025 (2025), p. 269. DOI: 10.22323/1.501.0269.

[4] Doug R. Bergman and Isaac Buckland. "The Cosmic Ray Event Spectrum in the Knee Region measured by the NICHE Array at Telescope Array". In: *PoS* ICRC2025 (2025), p. 188. DOI: 10.22323/1.501.0188.

[5] Douglas R. Bergman et al. "Updated comparison of the UHECR energy spectra measured by the Pierre Auger Observatory and the Telescope Array". In: *PoS* ICRC2025 (2025), p. 381. DOI: 10.22323/1.501.0381. arXiv: 2509.05530 [astro-ph.HE].

[6] Adila Abdul Halim et al. "Update on full-sky searches for large- and medium-scale anisotropies in the UHECR flux using the Pierre Auger Observatory and the Telescope Array". In: *PoS* UHECR2024 (2025), p. 009. DOI: 10.22323/1.484.0009.

[7] Jihyun Kim and Douglas Bergman. "The fractional analysis of mass composition measured by the Telescope Array FADC fluorescence detectors in hybrid mode". In: *PoS* UHECR2024 (2025), p. 098. DOI: 10.22323/1.484.0098.

[8] N. Kieu et al. "First Time-Resolved Leader Spectra Associated With a Downward Terrestrial Gamma-Ray Flash Detected at the Telescope Array Surface Detector". In: *J. Geophys. Res. Atmos.* 129.24 (2024), e2024JD041720. DOI: 10.1029/2024JD041720.

[9] R. U. Abbasi et al. "Intermediate Fluence Downward Terrestrial Gamma Ray Flashes as Observed by the Telescope Array Surface Detector". In: *J. Geophys. Res. Atmos.* 129.24 (2024), e2024JD041260. DOI: 10.1029/2024JD041260.

[10] R. U. Abbasi et al. "Isotropy of Cosmic Rays beyond 1020 eV Favors Their Heavy Mass Composition". In: *Phys. Rev. Lett.* 133.4 (2024), p. 041001. DOI: 10.1103/PhysRevLett.133.041001. arXiv: 2406.19287 [astro-ph.HE].

[11] R. U. Abbasi et al. "Mass composition of ultrahigh energy cosmic rays from distribution of their arrival directions with the Telescope Array". In: *Phys. Rev. D*

110.2 (2024), p. 022006. DOI: 10.1103/PhysRevD.110.022006. arXiv: 2406.19286 [astro-ph.HE].

[12] R. U. Abbasi et al. "Observation of Declination Dependence in the Cosmic Ray Energy Spectrum". In: (June 2024). arXiv: 2406.08612 [astro-ph.HE].

[13] R. U. Abbasi et al. "An extremely energetic cosmic ray observed by a surface detector array". In: *Science* 382.6673 (2023), abo5095. DOI: 10.1126/science.abo5095. arXiv: 2311.14231 [astro-ph.HE].

[14] Jihyun Kim et al. "Highlights from the Telescope Array Experiment". In: *PoS* ICRC2023 (2024), p. 008. DOI: 10.22323/1.444.0008.

[15] Douglas R. Bergman et al. "Measurement of UHECR energy spectrum with the Pierre Auger Observatory and the Telescope Array". In: *PoS* ICRC2023 (2024), p. 406. DOI: 10.22323/1.444.0406.

[16] Adila Abdul Halim et al. "Possible interpretations of the joint observations of UHECR arrival directions using data recorded at the Telescope Array and the Pierre Auger Observatory". In: *PoS* ICRC2023 (2023), p. 528. DOI: 10.22323/1.444.0528.

[17] Frank Schroeder et al. "Snowmass UHECR Whitepaper: Requirements on Future Instrumentation". In: *PoS* ICRC2023 (2023), p. 206. DOI: 10.22323/1.444.0206.

[18] Isaac Buckland and Douglas Bergman. "CHASM (CHerenkov Air Shower Model)". In: *PoS* ICRC2023 (2023), p. 325. DOI: 10.22323/1.444.0325.

[19] Diksha Garg et al. "Neutrino propagation through Earth: modeling uncertainties using nuPyProp". In: *PoS* ICRC2023 (2023), p. 1115. DOI: 10.22323/1.444.1115. arXiv: 2308.13659 [hep-ph].

[20] R. U. Abbasi et al. "The energy spectrum of cosmic rays measured by the Telescope Array using 10 years of fluorescence detector data". In: *Astropart. Phys.* 151 (2023), p. 102864. DOI: 10.1016/j.astropartphys.2023.102864.

[21] D. R. Bergman et al. "The energy spectrum of ultra-high energy cosmic rays measured at the Pierre Auger Observatory and the Telescope Array". In: *EPJ Web Conf.* 283 (2023), p. 02003. DOI: 10.1051/epjconf/202328302003.

[22] Isaac J. Buckland and Douglas Bergman. "Universality of Cherenkov light in EAS". in: *Astropart. Phys.* 150 (2023), p. 102832. DOI: 10.1016/j.astropartphys.2023.102832. arXiv: 2303.15653 [astro-ph.HE].

[23] Douglas Bergman et al. "Testing the Compatibility of the Depth of the Shower Maximum Measurements performed at Telescope Array and the Pierre Auger Observatory - Auger-TA Mass Composition Working Group Report". In: *EPJ Web Conf.* 283 (2023), p. 02008. DOI: 10.1051/epjconf/202328302008.

[24] Diksha Garg et al. "Neutrino propagation in the Earth and emerging charged leptons with nuPyProp". In: *JCAP* 01 (2023), p. 041. DOI: 10.1088/1475-7516/2023/01/041. arXiv: 2209.15581 [astro-ph.HE].

[25] A. Coleman et al. "Ultra high energy cosmic rays The intersection of the Cosmic and Energy Frontiers". In: *Astropart. Phys.* 149 (2023), p. 102819. DOI: 10.1016/j.astropartphys.2023.102819. arXiv: 2205.05845 [astro-ph.HE].

[26] R. U. Abbasi et al. "First High-speed Video Camera Observations of a Lightning Flash Associated with a Downward Terrestrial Gamma-ray Flash". In: *Geophys. Res.*

Lett. 50 (May 2022), e2023. DOI: 10.1029/2023GL102958. arXiv: 2205.05115 [physics.ao-ph].

[27] Rasha Abbasi et al. "Highlights from the Telescope Array experiment". In: *PoS* ICRC2021 (2022), p. 012. DOI: 10.22323/1.395.0012.

[28] A. Albert et al. "Search for Spatial Correlations of Neutrinos with Ultra-high-energy Cosmic Rays". In: *Astrophys. J.* 934.2 (2022), p. 164. DOI: 10.3847/1538-4357/ac6def. arXiv: 2201.07313 [astro-ph.HE].

[29] Peter Tinyakov et al. "The UHECR dipole and quadrupole in the latest data from the original Auger and TA surface detectors". In: *PoS* ICRC2021 (2021), p. 375. DOI: 10.22323/1.395.0375. arXiv: 2111.14593 [astro-ph.HE].

[30] Armando di Matteo et al. "UHECR arrival directions in the latest data from the original Auger and TA surface detectors and nearby galaxies". In: *PoS* ICRC2021 (2021), p. 308. DOI: 10.22323/1.395.0308. arXiv: 2111.12366 [astro-ph.HE].

[31] R. U. Abbasi et al. "Observation of variations in cosmic ray single count rates during thunderstorms and implications for large-scale electric field changes". In: *Phys. Rev. D* 105.6 (2022), p. 062002. DOI: 10.1103/PhysRevD.105.062002. arXiv: 2111.09962 [astro-ph.HE].

[32] R. U. Abbasi et al. "Indications of a Cosmic Ray Source in the Perseus-Pisces Supercluster". In: (Oct. 2021). arXiv: 2110.14827 [astro-ph.HE].

[33] Sameer Patel et al. "Monte Carlo simulations of neutrino and charged lepton propagation in the Earth with nuPyProp". In: *PoS* ICRC2021 (2021), p. 1203. DOI: 10.22323/1.395.1203. arXiv: 2109.08198 [hep-ph].

[34] R. U. Abbasi et al. "Surface detectors of the TAx4 experiment". In: *Nucl. Instrum. Meth. A* 1019 (2021), p. 165726. DOI: 10.1016/j.nima.2021.165726. arXiv: 2103.01086 [astro-ph.IM].

[35] Isaac Buckland and Douglas Bergman. "CHASM (CHerenkov Air Shower Model): Simulating the Cherenkov Profiles of Cosmic Ray Air Showers". In: *PoS* ICRC2021 (2021), p. 234. DOI: 10.22323/1.395.0234.

[36] Rasha Abbasi et al. "Telescope Array search for EeV photons". In: *PoS* ICRC2021 (2021), p. 864. DOI: 10.22323/1.395.0864.

[37] Kazumasa Kawata et al. "Effects of Galactic magnetic field on the UHECR anisotropy studies". In: *PoS* ICRC2021 (2021), p. 358. DOI: 10.22323/1.395.0358.

[38] Yugo Omura et al. "Energy spectrum and the shower maxima of cosmic rays above the knee region measured with the NICHE detectors at the TA site". In: *PoS* ICRC2021 (2021), p. 329. DOI: 10.22323/1.395.0329.

[39] Hyomin Jeong et al. "Reconstruction of Air Shower Events Measured by the Surface Detectors of the TAx4 Experiment". In: *PoS* ICRC2021 (2021), p. 331. DOI: 10.22323/1.395.0331.

[40] Tareq AbuZayyad et al. "Cosmic Ray Energy Spectrum measured by the TALE Fluorescence Detector". In: *PoS* ICRC2021 (2021), p. 347. DOI: 10.22323/1.395.0347.

[41] Tareq AbuZayyad et al. "Cosmic Ray Composition between 2 PeV and 2 EeV measured by the TALE Fluorescence Detector". In: *PoS* ICRC2021 (2021), p. 346. DOI: 10.22323/1.395.0346.

[42] Rasha Abbasi et al. "Joint analysis of the energy spectrum of ultra-high-energy cosmic rays as measured at the Pierre Auger Observatory and the Telescope Array". In: *PoS* ICRC2021 (2021), p. 337. DOI: 10.22323/1.395.0337.

[43] Mikhail Kuznetsov et al. "UHECR mass composition from anisotropy of their arrival directions with the Telescope Array SD". in: *PoS* ICRC2021 (2021), p. 294. DOI: 10.22323/1.395.0294.

[44] Rasha Abbasi et al. "The measurements of the cosmic ray energy spectrum and the depth of maximum shower development of Telescope Array Hybrid trigger events". In: *PoS* ICRC2021 (2021), p. 305. DOI: 10.22323/1.395.0305.

[45] Takayuki Tomida et al. "The atmospheric transparency of Telescope Array observation site by the CLF". in: *PoS* ICRC2021 (2021), p. 217. DOI: 10.22323/1.395.0217.

[46] Rasha Abbasi et al. "A study of analysis method for the identification of UHECR source type". In: *PoS* ICRC2021 (2021), p. 242. DOI: 10.22323/1.395.0242.

[47] Rasha Abbasi et al. "Current status and prospects of surface detector of the TAx4 experiment". In: *PoS* ICRC2021 (2021), p. 203. DOI: 10.22323/1.395.0203.

[48] Rasha Abbasi et al. "Mass composition of Telescope Array's surface detectors events using deep learning". In: *PoS* ICRC2021 (2021), p. 384. DOI: 10.22323/1.395.0384.

[49] Dmitri Ivanov et al. "Recent measurement of the Telescope Array energy spectrum and observation of the shoulder feature in the Northern Hemisphere". In: *PoS* ICRC2021 (2021), p. 341. DOI: 10.22323/1.395.0341.

[50] Rasha Abbasi et al. "Cosmic ray energy spectrum in the 2nd knee region measured by the TALE-SD array". In: *PoS* ICRC2021 (2021), p. 362. DOI: 10.22323/1.395.0362.

[51] Rasha Abbasi et al. "Cosmic Ray Composition in the Second Knee Region as Measured by the TALE Hybrid Detector". In: *PoS* ICRC2021 (2021), p. 353. DOI: 10.22323/1.395.0353.

[52] Rasha Abbasi et al. "The status of the TALE surface detector array and TALE infill project". In: *PoS* ICRC2021 (2021), p. 255. DOI: 10.22323/1.395.0255.

[53] Toshiyuki Nonaka et al. "Study on the cosmic ray intensity variation using scintillation counters for air shower observation". In: *PoS* ICRC2021 (2021), p. 1255. DOI: 10.22323/1.395.1255.

[54] Douglas Bergman and Greg Furlich. "Telescope Array 10-Year Monocular Spectrum Measurement". In: *PoS* ICRC2021 (2021), p. 339. DOI: 10.22323/1.395.0339.

[55] Sangwoo Kim et al. "Analysis of TAx4 hybrid trigger and events". In: *PoS* ICRC2021 (2021), p. 332. DOI: 10.22323/1.395.0332.

[56] R. U. Abbasi et al. "Performance and simulation of the surface detector array of the TAx4 experiment". In: *PoS* ICRC2021 (2021), p. 355. DOI: 10.22323/1.395.0355.

[57] Rasha Abbasi et al. "FOV direction and image size calibration of Fluorescence Detector using light source on UAV". in: *PoS* ICRC2021 (2021), p. 196. DOI: 10.22323/1.395.0196.

[58] Douglas Bergman. "Telescope Array Combined Fit to Cosmic Ray Spectrum and Composition". In: *PoS* ICRC2021 (2021), p. 338. DOI: 10.22323/1.395.0338.

[59] Jörg R. Hörandel et al. "GCOS - The Global Cosmic Ray Observatory". In: *PoS* ICRC2021 (2021), p. 027. DOI: 10.22323/1.395.0027. arXiv: 2203.01127 [astro-ph.HE].

[60] R. U. Abbasi et al. "The Cosmic-Ray Composition between 2 PeV and 2 EeV Observed with the TALE Detector in Monocular Mode". In: *Astrophys. J.* 909.2 (2021), p. 178. DOI: 10.3847/1538-4357/abdd30. arXiv: 2012.10372 [astro-ph.HE].

[61] A. V. Olinto et al. "The POEMMA (Probe of Extreme Multi-Messenger Astrophysics) observatory". In: *JCAP* 06 (2021), p. 007. DOI: 10.1088/1475-7516/2021/06/007. arXiv: 2012.07945 [astro-ph.IM].

[62] T. Abu-Zayyad et al. "Observations of the Origin of Downward Terrestrial Gamma-Ray Flashes". In: (Sept. 2020). DOI: 10.1029/2019JD031940. arXiv: 2009.14327 [physics.ao-ph].

[63] A. V. Olinto et al. "Snowmass 2021 Letter of Interest: The Probe Of Multi-Messenger Astrophysics (POEMMA)". in: *Snowmass 2021*. Sept. 2020. arXiv: 2008.13047 [astro-ph.HE].

[64] Andrés Romero-Wolf et al. "Radio Detection of Ultra-high Energy Cosmic Rays with Low Lunar Orbiting SmallSats". In: (Aug. 2020). arXiv: 2008.11232 [astro-ph.HE].

[65] R. U. Abbasi et al. "Search for Large-scale Anisotropy on Arrival Directions of Ultra-high-energy Cosmic Rays Observed with the Telescope Array Experiment". In: *Astrophys. J. Lett.* 898.2 (2020), p. L28. DOI: 10.3847/2041-8213/aba0bc. arXiv: 2007.00023 [astro-ph.HE].

[66] R. U. Abbasi et al. "Measurement of the proton-air cross section with Telescope Array's Black Rock Mesa and Long Ridge fluorescence detectors, and surface array in hybrid mode". In: *Phys. Rev. D* 102.6 (2020), p. 062004. DOI: 10.1103/PhysRevD.102.062004. arXiv: 2006.05012 [astro-ph.HE].

[67] R. U. Abbasi et al. "Evidence for a Supergalactic Structure of Magnetic Deflection Multiplets of Ultra-High Energy Cosmic Rays". In: *Astrophys. J.* 899.1 (2020), p. 86. DOI: 10.3847/1538-4357/aba26c. arXiv: 2005.07312 [astro-ph.HE].

[68] A. Anker et al. "White Paper: ARIANNA-200 high energy neutrino telescope". In: (Apr. 2020). arXiv: 2004.09841 [astro-ph.IM].

[69] Douglas Bergman. "Telescope Array: Latest Results and Expansion Plans". In: *J. Phys. Conf. Ser.* 1468.1 (2020). Ed. by Masayuki Nakahata, p. 012078. DOI: 10.1088/1742-6596/1468/1/012078.

[70] Yugo Omura et al. "NICHE detector and operations". In: *J. Phys. Conf. Ser.* 1468.1 (2020). Ed. by Masayuki Nakahata, p. 012097. DOI: 10.1088/1742-6596/1468/1/012097.

[71] John F. Krizmanic et al. "νSpaceSim: A Comprehensive Neutrino Simulation Package for Space-based & Suborbital Experiments". In: *PoS* ICRC2019 (2020), p. 936. DOI: 10.22323/1.358.0936.

[72] Douglas Bergman et al. "First Results from NICHE and the NICHE-TALE Hybrid Detector". In: *PoS* ICRC2019 (2020), p. 189. DOI: 10.22323/1.358.0189.

[73] Douglas Bergman. "Combined Fit of the Spectrum and Composition from Telescope Array". In: *PoS* ICRC2019 (2020), p. 190. DOI: 10.22323/1.358.0190.

[74] Douglas Bergman and Thomas Stroman. "TA 10 Year Stereo Composition Measurement". In: *PoS* ICRC2019 (2020), p. 191. DOI: 10.22323/1.358.0191.

[75] Greg Furlich and Douglas R. Bergman. "Towards a Telescope Array 10 Year FD Monocular Energy Spectrum". In: *PoS* ICRC2019 (2020), p. 260. DOI: 10.22323/1.358.0260.

[76] Yugo Omura et al. "NICHE detector and operations". In: *PoS* ICRC2019 (2020), p. 379. DOI: 10.22323/1.358.0379.

[77] Angela V. Olinto et al. "The POEMMA (Probe of Extreme Multi-Messenger Astrophysics) mission". In: *PoS* ICRC2019 (2020), p. 378. DOI: 10.22323/1.358.0378. arXiv: 1909.09466 [astro-ph.IM].

[78] A. V. Olinto et al. "POEMMA (Probe of Extreme Multi-Messenger Astrophysics) design". In: (July 2019). arXiv: 1907.06217 [astro-ph.HE].

[79] Luis A. Anchordoqui et al. "Performance and science reach of the Probe of Extreme Multimessenger Astrophysics for ultrahigh-energy particles". In: *Phys. Rev. D* 101.2 (2020), p. 023012. DOI: 10.1103/PhysRevD.101.023012. arXiv: 1907.03694 [astro-ph.HE].

[80] R. U. Abbasi et al. "Search for Ultra-High-Energy Neutrinos with the Telescope Array Surface Detector". In: *J. Exp. Theor. Phys.* 131.2 (2020), pp. 255–264. DOI: 10.31857/S0044451020080052. arXiv: 1905.03738 [astro-ph.HE].

[81] R. U. Abbasi et al. "Search for point sources of ultra-high-energy photons with the Telescope Array surface detector". In: *Mon. Not. Roy. Astron. Soc.* 492.3 (2020), pp. 3984–3993. DOI: 10.1093/mnras/stz3618. arXiv: 1904.00300 [astro-ph.HE].

[82] Frank G. Schröder et al. "High-Energy Galactic Cosmic Rays (Astro2020 Science White Paper)". In: *Bull. Am. Astron. Soc.* 51 (Mar. 2019), p. 131. arXiv: 1903.07713 [astro-ph.HE].

[83] Fred Sarazin et al. "What is the nature and origin of the highest-energy particles in the universe?" In: *Bull. Am. Astron. Soc.* 51.3 (2019), p. 93. arXiv: 1903.04063 [astro-ph.HE].

[84] Douglas R. Bergman et al. "NICHE: Air-Cherenkov observation at the TA site". In: *EPJ Web Conf.* 210 (2019). Ed. by I. Lhenry-Yvon et al., p. 05001. DOI: 10.1051/epjconf/201921005001.

[85] R. U. Abbasi et al. "Constraints on the diffuse photon flux with energies above 10^{18} eV using the surface detector of the Telescope Array experiment". In: *Astropart. Phys.* 110 (2019), pp. 8–14. DOI: 10.1016/j.astropartphys.2019.03.003. arXiv: 1811.03920 [astro-ph.HE].

[86] R. U. Abbasi et al. "Testing a Reported Correlation between Arrival Directions of Ultra-high-energy Cosmic Rays and a Flux Pattern from nearby Starburst Galaxies using Telescope Array Data". In: *Astrophys. J. Lett.* 867.2 (2018), p. L27. DOI: 10.3847/2041-8213/aaebf9. arXiv: 1809.01573 [astro-ph.HE].

[87] R. U. Abbasi et al. "Mass composition of ultrahigh-energy cosmic rays with the Telescope Array Surface Detector data". In: *Phys. Rev. D* 99.2 (2019), p. 022002. DOI: 10.1103/PhysRevD.99.022002. arXiv: 1808.03680 [astro-ph.HE].

[88] R. U. Abbasi et al. "Study of muons from ultrahigh energy cosmic ray air showers measured with the Telescope Array experiment". In: *Phys. Rev. D* 98.2 (2018), p. 022002. DOI: 10.1103/PhysRevD.98.022002. arXiv: 1804.03877 [astro-ph.HE].

[89] R. U. Abbasi et al. "The Cosmic-Ray Energy Spectrum between 2 PeV and 2 EeV Observed with the TALE detector in monocular mode". In: *Astrophys. J.* 865.1 (2018), p. 74. DOI: 10.3847/1538-4357/aada05. arXiv: 1803.01288 [astro-ph.HE].

[90] R. U. Abbasi et al. "Evidence of Intermediate-Scale Energy Spectrum Anisotropy of Cosmic Rays E$\geq 10^{19.2}$ eV with the Telescope Array Surface Detector". In: *Astrophys. J.* 862.2 (2018), p. 91. DOI: 10.3847/1538-4357/aac9c8. arXiv: 1802.05003 [astro-ph.HE].

[91] R. U. Abbasi et al. "Depth of Ultra High Energy Cosmic Ray Induced Air Shower Maxima Measured by the Telescope Array Black Rock and Long Ridge FADC Fluorescence Detectors and Surface Array in Hybrid Mode". In: *Astrophys. J.* 858.2 (2018), p. 76. DOI: 10.3847/1538-4357/aabad7. arXiv: 1801.09784 [astro-ph.HE].

[92] R. U. Abbasi et al. "Evidence for Declination Dependence of the Ultrahigh Energy Cosmic Ray Spectrum in the Northern Hemisphere". In: (Jan. 2018). arXiv: 1801.07820 [astro-ph.HE].

[93] Yoshiki Tsunesada et al. "NICHE: Non-Imaging Cherenkov Light Observation at the TA Site". In: *JPS Conf. Proc.* 19 (2018). Ed. by K. Kawata et al., p. 011042. DOI: 10.7566/JPSCP.19.011042.

[94] R. U. Abbasi et al. "The bursts of high energy events observed by the telescope array surface detector". In: *Phys. Lett. A* 381.32 (2017), pp. 2565–2572. DOI: 10.1016/j.physleta.2017.06.022.

[95] Angela V. Olinto et al. "POEMMA: Probe Of Extreme Multi-Messenger Astrophysics". In: *PoS* ICRC2017 (2018), p. 542. DOI: 10.22323/1.301.0542. arXiv: 1708.07599 [astro-ph.IM].

[96] M. Abe et al. "Anisotropy search in the Ultra High Energy Cosmic Ray Spectrum in the Northern Hemisphere using latest data obtained with Telescope Array surface detector". In: *PoS* ICRC2021 (2021), p. 344. DOI: 10.22323/1.395.0344. arXiv: 1707.04967 [astro-ph.HE].

[97] Douglas Bergman et al. "jNICHE: Prototype detectors of a non-imaging Cherenkov array at the TA site". In: *PoS* ICRC2017 (2018), p. 415. DOI: 10.22323/1.301.0415.

[98] Douglas Bergman and Thomas Stroman. "Telescope Array measurement of UHECR composition from stereoscopic fluorescence detection". In: *PoS* ICRC2017 (2018), p. 538. DOI: 10.22323/1.301.0538.

[99] R. U. Abbasi et al. "Gamma-ray Showers Observed at Ground Level in Coincidence With Downward Lightning Leaders". In: *J. Geophys. Res. Atmos.* 123 (2018), p. 6864. DOI: 10.1029/2017JD027931. arXiv: 1705.06258 [physics.ao-ph].

[100] R. U. Abbasi et al. "Search for EeV Protons of Galactic Origin". In: *Astropart. Phys.* 86 (2017), pp. 21–26. DOI: 10.1016/j.astropartphys.2016.11.001. arXiv: 1608.06306 [astro-ph.HE].

[101] R. U. Abbasi et al. "First Upper Limits on the Radar Cross Section of Cosmic-Ray Induced Extensive Air Showers". In: *Astropart. Phys.* 87 (2017), pp. 1–17. DOI: 10.1016/j.astropartphys.2016.11.006. arXiv: 1603.05217 [astro-ph.IM].

[102] M. G. Aartsen et al. "Search for correlations between the arrival directions of IceCube neutrino events and ultrahigh-energy cosmic rays detected by the Pierre Auger Observatory and the Telescope Array". In: *JCAP* 01 (2016), p. 037. DOI: 10.1088/1475-7516/2016/01/037. arXiv: 1511.09408 [astro-ph.HE].

[103] R. U. Abbasi et al. "The energy spectrum of cosmic rays above $10^{17.2}$ eV measured by the fluorescence detectors of the Telescope Array experiment in seven years". In: *Astropart. Phys.* 80 (2016), pp. 131–140. DOI: 10.1016/j.astropartphys.2016.04.002. arXiv: 1511.07510 [astro-ph.HE].

[104] R. U. Abbasi et al. "Pierre Auger Observatory and Telescope Array: Joint Contributions to the 34th International Cosmic Ray Conference (ICRC 2015)". In: (Nov. 2015). arXiv: 1511.02103 [astro-ph.HE].

[105] M. G. Aartsen et al. "The IceCube Neutrino Observatory, the Pierre Auger Observatory and the Telescope Array: Joint Contribution to the 34th International Cosmic Ray Conference (ICRC 2015)". In: (Nov. 2015). arXiv: 1511.02109 [astro-ph.HE].

[106] John Krizmanic, Douglas Bergman, and Yoshiki Tsunesada. "The Cosmic Ray Nuclear Composition Measurement Performance of the Non-Imaging CHErenkov Array (NICHE)". in: *PoS* ICRC2015 (2016), p. 562. DOI: 10.22323/1.236.0562.

[107] Douglas R. Bergman and Thomas A. Stroman. "The distribution of shower longitudinal profile widths as measured by Telescope Array in stereo mode". In: *PoS* ICRC2015 (2016), p. 339. DOI: 10.22323/1.236.0339.

[108] Douglas Bergman and Tareq Abuzayyad. "Imaging and non-imaging Cherenkov hybrid reconstruction". In: *PoS* ICRC2015 (2016), p. 345. DOI: 10.22323/1.236.0345.

[109] Douglas R. Bergman, John F. Krizmanic, and Yoshiki Tsunesada. "The NICHE Array: status and plans". In: *PoS* ICRC2015 (2016), p. 635. DOI: 10.22323/1.236.0635.

[110] R. U. Abbasi et al. "The hybrid energy spectrum of Telescope Array's Middle Drum Detector and surface array". In: *Astropart. Phys.* 68 (2015), pp. 27–44. DOI: 10.1016/j.astropartphys.2015.02.008.

[111] R. U. Abbasi et al. "Measurement of the proton-air cross section with Telescope Array's Middle Drum detector and surface array in hybrid mode". In: *Phys. Rev. D* 92.3 (2015), p. 032007. DOI: 10.1103/PhysRevD.92.032007. arXiv: 1505.01860 [astro-ph.HE].

[112] B. K. Shin et al. "Gain monitoring of telescope array photomultiplier cameras for the first 4 years of operation". In: *Nucl. Instrum. Meth. A* 768 (2014), pp. 96–103. DOI: 10.1016/j.nima.2014.09.059.

[113] Alexander Aab et al. "Searches for Large-Scale Anisotropy in the Arrival Directions of Cosmic Rays Detected above Energy of 10^{19} eV at the Pierre Auger Observatory and the Telescope Array". In: *Astrophys. J.* 794.2 (2014), p. 172. DOI: 10.1088/0004-637X/794/2/172. arXiv: 1409.3128 [astro-ph.HE].

[114] R. U. Abbasi et al. "Study of Ultra-High Energy Cosmic Ray composition using Telescope Array's Middle Drum detector and surface array in hybrid mode". In: *Astropart. Phys.* 64 (2015), pp. 49–62. DOI: 10.1016/j.astropartphys.2014.11.004. arXiv: 1408.1726 [astro-ph.HE].

[115] R. U. Abbasi et al. "A Northern Sky Survey for Point-Like Sources of EeV Neutral Particles with the Telescope Array Experiment". In: *Astrophys. J.* 804.2 (2015), p. 133. DOI: 10.1088/0004-637X/804/2/133. arXiv: 1407.6145 [astro-ph.HE].

[116] R. U. Abbasi et al. "Indications of Intermediate-Scale Anisotropy of Cosmic Rays with Energy Greater Than 57 EeV in the Northern Sky Measured with the Surface Detector of the Telescope Array Experiment". In: *Astrophys. J. Lett.* 790 (2014), p. L21. DOI: 10.1088/2041-8205/790/2/L21. arXiv: 1404.5890 [astro-ph.HE].

[117] T. Abu-Zayyad et al. "CORSIKA Simulation of the Telescope Array Surface Detector". In: (Mar. 2014). arXiv: 1403.0644 [astro-ph.IM].

[118] T. Abu-Zayyad et al. "Pierre Auger Observatory and Telescope Array: Joint Contributions to the 33rd International Cosmic Ray Conference (ICRC 2013)". In: *33rd International Cosmic Ray Conference*. Oct. 2013. arXiv: 1310.0647 [astro-ph.HE].

[119] John Krizmanic, Douglas Bergman, and Pierre Sokolsky. "The Non-Imaging CHErenkov Array (NICHE): A TA/TALE Extension to Measure the Flux and Composition of Very-High Energy Cosmic Rays". In: *33rd International Cosmic Ray Conference*. 2013, p. 0365. arXiv: 1307.3912 [astro-ph.IM].

[120] John Krizmanic, Douglas Bergman, and Pierre Sokolsky. "The modeling of the nuclear composition measurement performance of the Non-Imaging CHErenkov Array (NICHE)". in: *33rd International Cosmic Ray Conference*. July 2013, p. 0366. arXiv: 1307.3918 [astro-ph.IM].

[121] T. Abu-Zayyad et al. "Correlations of the Arrival Directions of Ultra-high Energy Cosmic Rays with Extragalactic Objects as Observed by the Telescope Array Experiment". In: *Astrophys. J.* 777 (2013), p. 88. DOI: 10.1088/0004-637X/777/2/88. arXiv: 1306.5808 [astro-ph.HE].

[122] T. Abu-Zayyad et al. "Energy Spectrum of Ultra-High Energy Cosmic Rays Observed with the Telescope Array Using a Hybrid Technique". In: *Astropart. Phys.* 61 (2015), pp. 93–101. DOI: 10.1016/j.astropartphys.2014.05.002. arXiv: 1305.7273 [astro-ph.HE].

[123] T. Abu-Zayyad et al. "The Energy Spectrum of Ultra-High-Energy Cosmic Rays Measured by the Telescope Array FADC Fluorescence Detectors in Monocular Mode". In: *Astropart. Phys.* 48 (2013), pp. 16–24. DOI: 10.1016/j.astropartphys.2013.06.007. arXiv: 1305.6079 [astro-ph.HE].

[124] T. Abu-Zayyad et al. "Upper limit on the flux of photons with energies above 10^{19} eV using the Telescope Array surface detector". In: *Phys. Rev. D* 88.11 (2013), p. 112005. DOI: 10.1103/PhysRevD.88.112005. arXiv: 1304.5614 [astro-ph.HE].

[125] Douglas R. Bergman. "HiRes and TA spectrum measurements". In: *EPJ Web Conf.* 53 (2013). Ed. by K. -H. Kampert et al., p. 04001. DOI: 10.1051/epjconf/20135304001.

[126] Douglas Bergman. "TA Spectrum Summary". In: *33rd International Cosmic Ray Conference*. 2013, p. 0221.

[127] Thomas Stroman et al. "Monocular Measurement of the UHECR Energy Spectrum by the Telescope Array Fluorescence Detectors". In: *33rd International Cosmic Ray Conference*. 2013, p. 0476.

[128] Douglas Bergman. "An Efficient Technique for the Reconstruction of Extensive Air Showers using Non-Imaging Cherenkov Measurements". In: *33rd International Cosmic Ray Conference*. 2013, p. 0983.

[129] Douglas Bergman and John Krizmanic. "NICHE: The Non-Imaging CHErenkov Array". In: *AIP Conf. Proc.* 1516.1 (2013). Ed. by Jonathan F. Ormes, pp. 282–287. DOI: 10.1063/1.4792585. arXiv: 1212.6237 [astro-ph.IM].

[130] T. Abu-Zayyad et al. "The Cosmic Ray Energy Spectrum Observed with the Surface Detector of the Telescope Array Experiment". In: *Astrophys. J. Lett.* 768 (2013), p. L1. DOI: 10.1088/2041-8205/768/1/L1. arXiv: 1205.5067 [astro-ph.HE].

[131] T. Abu-Zayyad et al. "Search for Anisotropy of Ultra-High Energy Cosmic Rays with the Telescope Array Experiment". In: *Astrophys. J.* 757 (2012), p. 26. DOI: 10.1088/0004-637X/757/1/26. arXiv: 1205.5984 [astro-ph.HE].

[132] T. Abu-Zayyad et al. "The Energy Spectrum of Telescope Array's Middle Drum Detector and the Direct Comparison to the High Resolution Fly's Eye Experiment". In: *Astropart. Phys.* 39-40 (2012), pp. 109–119. DOI: 10.1016/j.astropartphys.2012.05.012. arXiv: 1202.5141 [astro-ph.IM].

[133] H. Tokuno et al. "New air fluorescence detectors employed in the Telescope Array experiment". In: *Nucl. Instrum. Meth. A* 676 (2012), pp. 54–65. DOI: 10.1016/j.nima.2012.02.044. arXiv: 1201.0002 [astro-ph.IM].

[134] T. Abu-Zayyad et al. "The surface detector array of the Telescope Array experiment". In: *Nucl. Instrum. Meth. A* 689 (2013), pp. 87–97. DOI: 10.1016/j.nima.2012.05.079. arXiv: 1201.4964 [astro-ph.IM].

[135] S. R. Stratton et al. "Using the Monte Carlo Technique in the Observation of Fluorescence from UHECRs". In: *32nd International Cosmic Ray Conference*. Vol. 2. Aug. 2011, p. 262. DOI: 10.7529/ICRC2011/V02/1299.

[136] D. R. Bergman et al. "The Energy Spectrum of UHECRs using the TA Fluorescence Detectors in Monocular Mode". In: *32nd International Cosmic Ray Conference*. Vol. 2. Aug. 2011, p. 265. DOI: 10.7529/ICRC2011/V02/1300.

[137] *Hadron-Hadron and Cosmic-Ray Interactions at multi-TeV Energies: Mini-proceedings, workshop, Trento, Italy, 29 Nov - 3 Dec 2010*. Jan. 2011. arXiv: 1101.1852 [hep-ex].

[138] H. Tokuno et al. "The status of the Telescope Array experiment". In: *J. Phys. Conf. Ser.* 293 (2011). Ed. by Yifang Wang, p. 012035. DOI: 10.1088/1742-6596/293/1/012035.

[139] T. A. Stroman et al. "Cross-calibration of Telescope Array Fluorescence Detectors with Static and Roving Standard Candles". In: *32nd International Cosmic Ray Conference*. Vol. 3. 2011, p. 333. DOI: 10.7529/ICRC2011/V03/1301.

[140] H. Tokuno et al. "The telescope array experiment: Status and prospects". In: *AIP Conf. Proc.* 1238.1 (2010). Ed. by Hajime Susa et al., pp. 365–368. DOI: 10.1063/1.3455968.

[141] R. U. Abbasi et al. "Analysis of Large-Scale Anisotropy of Ultra-High Energy Cosmic Rays in HiRes Stereo Data". In: *Astrophys. J. Lett.* 713.1 (2010), pp. L64–L68. DOI: 10.1088/2041-8205/713/1/L64. arXiv: 1002.1444 [astro-ph.HE].

[142] R. U. Abbasi et al. "Indications of Proton-Dominated Cosmic Ray Composition above 1.6 EeV". in: *Phys. Rev. Lett.* 104 (2010), p. 161101. DOI: 10.1103/PhysRevLett.104.161101. arXiv: 0910.4184 [astro-ph.HE].

[143] R. U. Abbasi et al. "Measurement of the Flux of Ultra High Energy Cosmic Rays by the Stereo Technique". In: *Astropart. Phys.* 32 (2009), pp. 53–60. DOI: 10.1016/j.astropartphys.2009.06.001. arXiv: 0904.4500 [astro-ph.HE].

[144] T. Nonaka et al. "The present status of the Telescope Array experiment". In: *Nucl. Phys. B Proc. Suppl.* 190 (2009). Ed. by R. Caruso et al., pp. 26–31. DOI: 10.1016/j.nuclphysbps.2009.03.064.

[145] D. R. Bergman. "Latest Results from HiRes". In: *20th Rencontres de Blois on Challenges in Particle Astrophysics.* 2008, pp. 45–54. arXiv: 0807.2814 [astro-ph].

[146] R. U. Abbasi et al. "Search for Correlations between HiRes Stereo Events and Active Galactic Nuclei". In: *Astropart. Phys.* 30 (2008), pp. 175–179. DOI: 10.1016/j.astropartphys.2008.08.004. arXiv: 0804.0382 [astro-ph].

[147] R. U. Abbasi et al. "An upper limit on the electron-neutrino flux from the HiRes detector". In: *Astrophys. J.* 684 (2008), pp. 790–793. DOI: 10.1086/590335. arXiv: 0803.0554 [astro-ph].

[148] H. Kawai et al. "Telescope array experiment". In: *Nucl. Phys. B Proc. Suppl.* 175-176 (2008). Ed. by Kwong Sang Cheng et al., pp. 221–226. DOI: 10.1016/j.nuclphysbps.2007.11.002.

[149] H. Tokuno et al. "The Telescope Array experiment: Status and prospects". In: *J. Phys. Conf. Ser.* 120 (2008). Ed. by Kunio Inoue, Atsuto Suzuki, and Tadao Mitsui, p. 062027. DOI: 10.1088/1742-6596/120/6/062027.

[150] R. Abbasi et al. "Techniques of the FLASH thin target experiment". In: *Nucl. Instrum. Meth. A* 597 (2008), pp. 32–36. DOI: 10.1016/j.nima.2008.08.054.

[151] R. Abbasi et al. "The FLASH thick-target experiment". In: *Nucl. Instrum. Meth. A* 597 (2008), pp. 37–40. DOI: 10.1016/j.nima.2008.08.053.

[152] R. Abbasi et al. "Air fluorescence measurements in the spectral range 300-420 nm using a 28.5-GeV electron beam". In: *Astropart. Phys.* 29 (2008), pp. 77–86. DOI: 10.1016/j.astropartphys.2007.11.010. arXiv: 0708.3116 [astro-ph].

[153] Yoshiki Tsunesada et al. "Telescope Array Aperture: Mono, Stereo and Hybrid". In: *30th International Cosmic Ray Conference.* Vol. 4. July 2007, pp. 409–412.

[154] Masaki Fukushima et al. "Status and Prospect of Telescope Array (TA) Experiment". In: *30th International Cosmic Ray Conference.* Vol. 4. 2008, pp. 417–420.

[155] Douglas Bergman. "The TALE Tower Detector". In: *30th International Cosmic Ray Conference.* Vol. 5. July 2007, pp. 1141–1144.

[156] Douglas R. Bergman. "Observation of the GZK Cutoff by the HiRes Experiment". In: *30th International Cosmic Ray Conference*. Vol. 4. July 2007, pp. 451–454.

[157] Olga Brusova, D. R. Bergman, and K. Martens. "Neutrino and Gamma Ray Flux Expectations from HiRes Monocular Fits". In: *30th International Cosmic Ray Conference*. Vol. 4. July 2007, pp. 551–554.

[158] Douglas R. Bergman and John W. Belz. "Cosmic Rays: The Second Knee and Beyond". In: *J. Phys. G* 34 (2007), R359. DOI: 10.1088/0954-3899/34/10/R01. arXiv: 0704.3721 [astro-ph].

[159] R. U. Abbasi et al. "First observation of the Greisen-Zatsepin-Kuzmin suppression". In: *Phys. Rev. Lett.* 100 (2008), p. 101101. DOI: 10.1103/PhysRevLett.100.101101. arXiv: astro-ph/0703099.

[160] Rasha Abbasi et al. "Alternative Methods to Finding Patterns in HiRes Stereo Data". In: *Astropart. Phys.* 28 (2007), pp. 385–389. DOI: 10.1016/j.astropartphys.2007.08.005. arXiv: astro-ph/0702361.

[161] Douglas R. Bergman. "Observation of the GZK Cutoff Using the HiRes Detector". In: *Nucl. Phys. B Proc. Suppl.* 165 (2007). Ed. by R. Caruso et al., pp. 19–26. DOI: 10.1016/j.nuclphysbps.2006.11.004. arXiv: astro-ph/0609453.

[162] R. Abbasi et al. "Studies of systematic uncertainties in the estimation of the monocular aperture of the HiRes experiment". In: *Astropart. Phys.* 27 (2007), pp. 370–381. DOI: 10.1016/j.astropartphys.2006.12.004. arXiv: astro-ph/0607094.

[163] R. Abbasi et al. "A Likelihood Method for Measuring the Ultrahigh Energy Cosmic Ray Composition". In: *Astropart. Phys.* 26 (2006), pp. 28–40. DOI: 10.1016/j.astropartphys.2006.04.007. arXiv: astro-ph/0604558.

[164] Douglas R. Bergman et al. "Can experiments studying ultrahigh energy cosmic rays measure the evolution of the sources?" In: (Mar. 2006). arXiv: astro-ph/0603797.

[165] R. Abbasi et al. "A measurement of time-averaged aerosol optical depth using air-showers observed in stereo by hires". In: *Astropart. Phys.* 25 (2006), pp. 93–97. DOI: 10.1016/j.astropartphys.2006.01.001. arXiv: astro-ph/0601091.

[166] D. Bergman. "Fitting the HiRes data". In: *J. Phys. Conf. Ser.* 47 (2006), pp. 154–159. DOI: 10.1088/1742-6596/47/1/019.

[167] R. Abbasi et al. "Techniques for measuring atmospheric aerosols at the high resolution fly's eye experiment". In: *Astropart. Phys.* 25 (2006), pp. 74–83. DOI: 10.1016/j.astropartphys.2005.11.006. arXiv: astro-ph/0512423.

[168] Katsuaki Kasahara et al. "The current status and prospect of the ta experiment". In: *11th International Conference on Elastic and Diffractive Scattering: Towards High Energy Frontiers: The 20th Anniversary of the Blois Workshops, 17th Rencontre de Blois*. Nov. 2005. arXiv: astro-ph/0511177.

[169] R. U. Abbasi et al. "Search for cross-correlations of ultrahigh-energy cosmic rays with BL Lacertae objects". In: *Astrophys. J.* 636 (2006), pp. 680–684. DOI: 10.1086/498142. arXiv: astro-ph/0507120.

[170] D. R. Bergman. "UHECR Composition Measurements Using the HiRes-II Detector". In: (July 2005). arXiv: astro-ph/0507483.

[171] D. R. Bergman. "Fitting the HiRes Spectra". In: (July 2005). arXiv: astro-ph/0507484.

[172] R. U. Abbasi et al. "Search for point-like sources of cosmic rays with energies above 10** 18.5 eV in the HiRes. 1. Monocular data-set". In: *Astropart. Phys.* 27 (2007), pp. 512–520. DOI: 10.1016/j.astropartphys.2007.03.002. arXiv: astro-ph/0507663.

[173] H. Ma et al. "First observation of the decay K+ —> e+ nu(e) mu+ mu-". In: *Phys. Rev. D* 73 (2006), p. 037101. DOI: 10.1103/PhysRevD.73.037101. arXiv: hep-ex/0505011.

[174] Aleksey Sher et al. "An Improved upper limit on the decay K+ —> pi+ mu+ e-". In: *Phys. Rev. D* 72 (2005), p. 012005. DOI: 10.1103/PhysRevD.72.012005. arXiv: hep-ex/0502020.

[175] R. U. Abbasi et al. "Observation of the ankle and evidence for a high-energy break in the cosmic ray spectrum". In: *Phys. Lett. B* 619 (2005), pp. 271–280. DOI: 10.1016/j.physletb.2005.05.064. arXiv: astro-ph/0501317.

[176] D. R. Bergman. "Fitting the HiRes spectra and monocular composition". In: *Int. J. Mod. Phys. A* 20 (2005). Ed. by John A. Ellison, pp. 3143–3146. DOI: 10.1142/S0217751X05025978.

[177] R. U. Abbasi et al. "Search for point sources of ultra-high energy cosmic rays above 4.0 x 10**19 eV using a maximum likelihood ratio test". In: *Astrophys. J.* 623 (2005), pp. 164–170. DOI: 10.1086/428570. arXiv: astro-ph/0412617.

[178] K. Reil et al. "Measurements of the fluorescence light yield in electromagnetic showers". In: *eConf* C041213 (2004). Ed. by P. Chen et al., p. 2417.

[179] Douglas R. Bergman. "Fitting the HiRes spectra and monocular composition". In: *Nucl. Phys. B Proc. Suppl.* 136 (2004). Ed. by C. Aramo, A. Insolia, and C. Tuve, pp. 40–45. DOI: 10.1016/j.nuclphysbps.2004.10.042. arXiv: astro-ph/0407244.

[180] R. U. Abbasi et al. "A Study of the composition of ultrahigh energy cosmic rays using the High Resolution Fly's Eye". In: *Astrophys. J.* 622 (2005), pp. 910–926. DOI: 10.1086/427931. arXiv: astro-ph/0407622.

[181] R. U. Abbasi et al. "Study of small - scale anisotropy of ultrahigh energy cosmic rays observed in stereo by HIRES". in: *Astrophys. J. Lett.* 610 (2004), p. L73. DOI: 10.1086/423303. arXiv: astro-ph/0404137.

[182] R. U. Abbasi et al. "A Search for arrival direction clustering in the HiRes-I monocular data above 10**19.5 eV". in: *Astropart. Phys.* 22 (2004), pp. 139–149. DOI: 10.1016/j.astropartphys.2004.07.003. arXiv: astro-ph/0404366.

[183] A. Sher et al. "New, high statistics measurement of the K+ —> pi0 e+ nu (K+(e3)) branching ratio". In: *AIP Conf. Proc.* 698.1 (2004). Ed. by Zohreh Parsa, pp. 381–384. DOI: 10.1063/1.1664262.

[184] R. Abbasi et al. "Search for global dipole enhancements in the HiRes-I monocular data above 10**18.5-eV". in: *Astropart. Phys.* 21 (2004), p. 111. DOI: 10.1016/j.astropartphys.2003.11.003. arXiv: astro-ph/0309457.

[185] Douglas R. Bergman. "Chasing the GZK with HiRes". In: *Mod. Phys. Lett. A* 18 (2003), pp. 1235–1245. DOI: 10.1142/S0217732303011010. arXiv: hep-ex/0307059.

[186] D. R. Bergman. "Measurement of the flux of UHE cosmic rays by the HiRes detectors observing in monocular mode". In: *28th International Cosmic Ray Conference*. July 2003, pp. 397–400.

[187] D. R. Bergman et al. "Fits of the HiRes spectrum to astrophysical models". In: *28th International Cosmic Ray Conference*. July 2003, pp. 683–686.

[188] A. Sher et al. "New, high statistics measurement of the K+ —> pi0 e+ nu (K+(e3)) branching ratio". In: *Phys. Rev. Lett.* 91 (2003), p. 261802. DOI: 10.1103/PhysRevLett.91.261802. arXiv: hep-ex/0305042.

[189] S. Pislak et al. "High statistics measurement of K(e4) decay properties". In: *Phys. Rev. D* 67 (2003). [Erratum: Phys.Rev.D 81, 119903 (2010)], p. 072004. DOI: 10.1103/PhysRevD.67.072004. arXiv: hep-ex/0301040.

[190] A. Alavi-Harati et al. "Search for the $K_L \to \pi^0\pi^0 e^+e^-$ Decay in the KTeV Experiment". In: *Phys. Rev. Lett.* 89 (2002), p. 211801. DOI: 10.1103/PhysRevLett.89.211801. arXiv: hep-ex/0210056.

[191] A. Alavi-Harati et al. "Measurements of Direct CP Violation, CPT Symmetry, and Other Parameters in the Neutral Kaon System". In: *Phys. Rev. D* 67 (2003). [Erratum: Phys.Rev.D 70, 079904 (2004)], p. 012005. DOI: 10.1103/PhysRevD.70.079904. arXiv: hep-ex/0208007.

[192] R. U. Abbasi et al. "Measurement of the flux of ultrahigh energy cosmic rays from monocular observations by the High Resolution Fly's Eye experiment". In: *Phys. Rev. Lett.* 92 (2004), p. 151101. DOI: 10.1103/PhysRevLett.92.151101. arXiv: astro-ph/0208243.

[193] R. U. Abbasi et al. "Monocular Measurement of the Spectrum of UHE Cosmic Rays by the FADC Detector of the HiRes Experiment". In: *Astropart. Phys.* 23 (2005), pp. 157–174. DOI: 10.1016/j.astropartphys.2004.12.006. arXiv: astro-ph/0208301.

[194] Douglas R. Bergman. "Monocular UHECR spectra as measured by HiRes". In: *Nucl. Phys. B Proc. Suppl.* 117 (2003). Ed. by S. Bentvelsen et al., p. 106. DOI: 10.1016/S0920-5632(03)90500-7. arXiv: hep-ex/0208024.

[195] A. A. Poblaguev et al. "Experimental study of the radiative decays K+ —> mu+ nu e+ e- and K+ —> e+ nu e+ e-". In: *Phys. Rev. Lett.* 89 (2002), p. 061803. DOI: 10.1103/PhysRevLett.89.061803. arXiv: hep-ex/0204006.

[196] A. Alavi-Harati et al. "A Measurement of the K_L Charge Asymmetry". In: *Phys. Rev. Lett.* 88 (2002), p. 181601. DOI: 10.1103/PhysRevLett.88.181601. arXiv: hep-ex/0202016.

[197] R. Appel et al. "A large acceptance, high-resolution detector for rare K+ decay experiments". In: *Nucl. Instrum. Meth. A* 479 (2002), pp. 349–406. DOI: 10.1016/S0168-9002(01)00906-8.

[198] P. A Sadowski et al. "Geometry and optics calibration for air fluorescence detectors using star light". In: *Astropart. Phys.* 18 (2002), pp. 237–248. DOI: 10.1016/S0927-6505(02)00143-3.

[199] A. Alavi-Harati et al. "Radiative Decay Width Measurements of Neutral Kaon Excitations using the Primakoff Effect". In: *Phys. Rev. Lett.* 89 (2002), p. 072001. DOI: 10.1103/PhysRevLett.89.072001. arXiv: hep-ex/0110016.

[200] A. Alavi-Harati et al. "Branching Ratio Measurement of the Decay $K_L \to e^+e^-\mu^+\mu^-$". In: *Phys. Rev. Lett.* 87 (2001), p. 111802. DOI: 10.1103/PhysRevLett.87.111802. arXiv: hep-ex/0108037.

[201] A. Alavi-Harati et al. "A New Measurement of the Radiative K_{e3} Branching Ratio and Photon Spectrum". In: *Phys. Rev. D* 64 (2001), p. 112004. DOI: 10.1103/PhysRevD.64.112004. arXiv: hep-ex/0106062.

[202] S. Pislak et al. "A New measurement of K+(e4) decay and the s wave pi pi scattering length a0(0)". In: *Phys. Rev. Lett.* 87 (2001). [Erratum: Phys.Rev.Lett. 105, 019901 (2010)], p. 221801. DOI: 10.1103/PhysRevLett.105.019901. arXiv: hep-ex/0106071.

[203] A. Alavi-Harati et al. "First Measurement of Form-Factors of the Decay $\Xi^0 \to \Sigma^+e^-\bar{\nu}_e$". In: *Phys. Rev. Lett.* 87 (2001), p. 132001. DOI: 10.1103/PhysRevLett.87.132001. arXiv: hep-ex/0105016.

[204] A. Alavi-Harati et al. "Measurements of the Rare Decay $K_L \to e^+e^-e^+e^-$". In: *Phys. Rev. Lett.* 86 (2001), pp. 5425–5429. DOI: 10.1103/PhysRevLett.86.5425. arXiv: hep-ex/0104043.

[205] A. Alavi-Harati et al. "Measurement of the Branching Ratio and Form-Factor of $K_L \to \mu^+\mu^-\gamma$". In: *Phys. Rev. Lett.* 87 (2001), p. 071801. DOI: 10.1103/PhysRevLett.87.071801.

[206] A. Alavi-Harati et al. "A Measurement of the Branching Ratio and Asymmetry of the Decay $\Xi^0 \to \Sigma^0\gamma$". In: *Phys. Rev. Lett.* 86 (2001), pp. 3239–3243. DOI: 10.1103/PhysRevLett.86.3239. arXiv: hep-ex/0012031.

[207] D. Bergman. "Recent results from the High Resolution Fly's Eye detector". In: *AIP Conf. Proc.* 586.1 (2001). Ed. by J. C. Wheeler and H. Martel, p. 827. DOI: 10.1063/1.1419666.

[208] A. Alavi-Harati et al. "First Observation of the Decay $K_L \to \pi^0e^+e^-\gamma$". In: *Phys. Rev. Lett.* 87 (2001), p. 021801. DOI: 10.1103/PhysRevLett.87.021801. arXiv: hep-ex/0011093.

[209] A. Alavi-Harati et al. "A Measurement of the Branching Ratio of $K_L \to e^+e^-\gamma\gamma$". In: *Phys. Rev. D* 64 (2001), p. 012003. DOI: 10.1103/PhysRevD.64.012003. arXiv: hep-ex/0010059.

[210] A. Alavi-Harati et al. "Search for the Decay $K_L \to \pi^0e^+e^-$". In: *Phys. Rev. Lett.* 86 (2001), pp. 397–401. DOI: 10.1103/PhysRevLett.86.397. arXiv: hep-ex/0009030.

[211] A. Alavi-Harati et al. "Study of the $K_L^0 \to \pi^+\pi^-\gamma$ Direct Emission Vertex". In: *Phys. Rev. Lett.* 86 (2001), pp. 761–764. DOI: 10.1103/PhysRevLett.86.761. arXiv: hep-ex/0008045.

[212] R. Appel et al. "Search for lepton flavor violation in K+ decays". In: *Phys. Rev. Lett.* 85 (2000), pp. 2877–2880. DOI: 10.1103/PhysRevLett.85.2877. arXiv: hep-ex/0006003.

[213] Douglas R Bergman. "A New measurement of the radiative K(e)3 branching ratio and photon spectrum". In: *Acta Phys. Polon. B* 31 (2000). Ed. by Lucjan Jarczyk et al., pp. 2133–2138. arXiv: hep-ex/0009032.

[214] R. Appel et al. "An Improved limit on the rate of decay K+ —> pi+ muon+ e-". In: *Phys. Rev. Lett.* 85 (2000), pp. 2450–2453. DOI: 10.1103/PhysRevLett.85.2450. arXiv: hep-ex/0005016.

[215] A. Alavi-Harati et al. "Observation of the Decay $K_L \to \mu^+\mu^-\gamma\gamma$". In: *Phys. Rev. D* 62 (2000), p. 112001. DOI: 10.1103/PhysRevD.62.112001. arXiv: hep-ex/0001005.

[216] A. Alavi-Harati et al. "Search for the Decay $K_L \to \pi^0\mu^+\mu^-$". In: *Phys. Rev. Lett.* 84 (2000), pp. 5279–5282. DOI: 10.1103/PhysRevLett.84.5279. arXiv: hep-ex/0001006.

[217] A. Alavi-Harati et al. "Search for the Weak Decay of a Lightly Bound H^0 Dibaryon". In: *Phys. Rev. Lett.* 84 (2000), pp. 2593–2597. DOI: 10.1103/PhysRevLett.84.2593. arXiv: hep-ex/9910030.

[218] H. Ma et al. "A New measurement of the rare decay K+ —> pi+ mu+ mu-". In: *Phys. Rev. Lett.* 84 (2000), pp. 2580–2583. DOI: 10.1103/PhysRevLett.84.2580. arXiv: hep-ex/9910047.

[219] A. Alavi-Harati et al. "Observation of CP Violation in $K_L \to \pi^+\pi^-e^+e^-$ Decays". In: *Phys. Rev. Lett.* 84 (2000), pp. 408–411. DOI: 10.1103/PhysRevLett.84.408. arXiv: hep-ex/9908020.

[220] A. Alavi-Harati et al. "Search for the Decay $K_L \to \pi^0\nu\bar{\nu}$ using $\pi^0- \to e^+e^-\gamma$". In: *Phys. Rev. D* 61 (2000), p. 072006. DOI: 10.1103/PhysRevD.61.072006. arXiv: hep-ex/9907014.

[221] R. Appel et al. "A New measurement of the properties of the rare decay K+ —> pi+ e+ e-". In: *Phys. Rev. Lett.* 83 (1999), pp. 4482–4485. DOI: 10.1103/PhysRevLett.83.4482. arXiv: hep-ex/9907045.

[222] A. Alavi-Harati et al. "Observation of Direct CP Violation in $K_{S,L} \to \pi\pi$ Decays". In: *Phys. Rev. Lett.* 83 (1999), pp. 22–27. DOI: 10.1103/PhysRevLett.83.22. arXiv: hep-ex/9905060.

[223] A. Alavi-Harati et al. "Measurement of the Branching Ratio of $\pi^0 \to e^+e^-$ using $K_L \to 3\pi^0$ Decays in Flight". In: *Phys. Rev. Lett.* 83 (1999), pp. 922–925. DOI: 10.1103/PhysRevLett.83.922. arXiv: hep-ex/9903007.

[224] A. Alavi-Harati et al. "Light Gluino Search for Decays Containing $\pi^+\pi^-$ or π^0 from a Neutral Hadron Beam at Fermilab". In: *Phys. Rev. Lett.* 83 (1999), pp. 2128–2132. DOI: 10.1103/PhysRevLett.83.2128. arXiv: hep-ex/9903048.

[225] A. Alavi-Harati et al. "Measurement of the Decay $K_L \to \pi^0\gamma\gamma$". In: *Phys. Rev. Lett.* 83 (1999), pp. 917–921. DOI: 10.1103/PhysRevLett.83.917. arXiv: hep-ex/9902029.

[226] Anthony A. Affolder et al. "Observation of the Decay $\Xi^0 \to \Sigma^+e^-\bar{\nu}_e$". In: *Phys. Rev. Lett.* 82 (1999), pp. 3751–3754. DOI: 10.1103/PhysRevLett.82.3751.

[227] Julia A. Thompson et al. "K+ —> pi+ mu+ mu- in E865 at BNL". in: *17th International Workshop on Weak Interactions and Neutrinos.* Jan. 1999, pp. 540–544. arXiv: hep-ex/9904026.

[228] J. Adams et al. "Search for the Decay $K_L \to \pi^0\nu\bar{\nu}$". In: *Phys. Lett. B* 447 (1999), pp. 240–245. DOI: 10.1016/S0370-2693(98)01593-7. arXiv: hep-ex/9806007.

[229] C. Bhat et al. "The CP/T Experiment: An Experimental Program to Study CP Violation and Search for CPT Violation in the $K_L - K_S$ System". In: (Apr. 1998).

[230] J. Adams et al. "Measurement of the Branching Fraction of the Decay $K_L \to \pi^+\pi^-e^+e^-$". In: *Phys. Rev. Lett.* 80 (1998), pp. 4123–4126. DOI: 10.1103/PhysRevLett.80.4123.

[231] A. Sher et al. "Recent results from BNL E865". In: *AIP Conf. Proc.* 459.1 (1999). Ed. by H. W. K. Cheung and J. N. Butler, pp. 431–440. DOI: 10.1063/1.57774.

[232] H. Ma et al. "Study of K+ –> pi+ e+ e- and K+ –> pi+ mu+ mu- decays in E865 at the AGS". in: *29th International Conference on High-Energy Physics.* 1998, pp. 1031–1035.

[233] T. Alexopoulos et al. "CP and CPT violation searches with K(S) mesons". In: *AIP Conf. Proc.* 412.1 (1997). Ed. by T. W. Donnelly, pp. 812–818. DOI: 10.1063/1.54372.

[234] Douglas Rolf Bergman. "A Search for the decay of a charged kaon into a pion, a muon and an electron". Other thesis. 1997.

[235] R. Appel et al. "Experimental studies of rare K+ decays". In: *28th International Conference on High-energy Physics.* July 1996, pp. 1188–1191.

[236] S. Pislak et al. "Rare K+ decays in flight". In: *Frascati Phys. Ser.* 7 (1997). Ed. by L. Koepke, pp. 407–414.